# Generative Modeling by Estimating Gradients of the Data Distribution

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# Executive Summary

This research paper explores **score-based generative modeling**, an emerging paradigm in machine learning that offers a principled alternative to traditional approaches like Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs). Instead of modeling the full data distribution or learning direct sampling procedures, score-based models estimate the **score function**—the gradient of the log-probability density of the data. This enables generation of new samples through techniques such as **Langevin dynamics** or **stochastic differential equations (SDEs)**.

The paper begins by contrasting score-based models with mainstream generative frameworks, emphasizing their theoretical simplicity and flexibility. It then introduces key methodological components, including **score matching**, **noise-conditioned training**, and **annealed Langevin dynamics**, highlighting how these techniques address the challenges of high-dimensional data generation.

A central focus is placed on the role of **SDE-based modeling**, which extends discrete noise schedules into a continuous-time framework. This approach improves sample quality, enables exact likelihood estimation, and offers a unified view of generative modeling that connects diffusion models, denoising autoencoders, and normalizing flows.

Despite their success, score-based models face challenges related to sampling efficiency, scalability, and applicability to discrete data. This paper concludes by identifying promising directions for future research, including faster samplers, better theoretical guarantees, and broader applications beyond image generation.

Overall, this work provides both a conceptual overview and practical insights into score-based generative modeling, offering a strong foundation for students and researchers interested in the next generation of AI-driven generative systems.

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## 1. Introduction and Overview

In the field of generative artificial intelligence, where the goal is to produce high-fidelity data such as images, audio, or text, two dominant paradigms have historically guided model design: **likelihood-based models** and **implicit generative models**.

**Likelihood-based models** aim to explicitly model the probability distribution of data. These models are trained using *maximum likelihood estimation* (MLE), which seeks parameters that maximize the probability of observed data under the model. Popular representatives include *autoregressive models* (e.g., GPT for language modeling), *normalizing flows*, and *variational autoencoders* (VAEs). While these models provide principled approaches to learning, they often impose architectural constraints to ensure tractability. For example, normalizing flows require invertible transformations with computable Jacobians, and VAEs rely on approximate inference techniques that may limit model expressiveness.

On the other hand, **implicit generative models** bypass the need to define an explicit probability distribution. Instead, they focus on learning a transformation from random noise to data samples, effectively capturing the data distribution through sampling. The most prominent example is the *Generative Adversarial Network* (GAN), which pits two networks—a generator and a discriminator—against each other in a game-theoretic framework. While GANs are capable of generating highly realistic outputs, they are notoriously difficult to train, often suffering from issues such as *mode collapse* (where only a limited variety of samples are produced) and training instability due to the adversarial dynamics.

Recent years have seen the emergence of a **third paradigm**: **score-based generative models**, also known as *score-based diffusion models* or *SDE-based models*. These models offer a compelling alternative by avoiding the pitfalls of both explicit likelihood modeling and adversarial training. Instead of modeling the full data distribution or generating samples directly, score-based models learn the *score function*—the gradient of the log-density of the data ($\nabla_x \log p(x)$). This function encapsulates the local structure of the data distribution and provides a direction that points toward regions of higher probability. Remarkably, by learning this gradient and using it in combination with stochastic sampling techniques, such as Langevin dynamics or stochastic differential equations (SDEs), score-based models are able to generate high-quality samples without needing to explicitly define or normalize p(x).

This paper explores the foundations, training methodology, and advances beyond naive score-based modeling. We aim to contrast them with traditional approaches, highlight their theoretical advantages, and discuss their growing impact in generative modeling tasks across various domains.

## 2. Methodology

This section presents the mathematical foundations of score-based generative models and the practical techniques used for training and sampling. It describes how the theoretical concept of score matching is operationalized into a scalable and effective generative modeling framework.

## 2.1. Score Functions and Matching Objectives

For any continuously differentiable probability density $p(x)$, the **score function** is defined as the gradient of the log-probability density with respect to the data variable $x$:

$$s(x) = \nabla_x \log p(x)$$

Conceptually, the score function represents a **vector field** over the data space. At any point $x$, the vector $s(x)$ points in the direction of steepest ascent of the log-density, effectively guiding samples toward regions of higher probability—i.e., the modes of the distribution. Importantly, the score function does not depend on the **normalizing constant** of $p(x)$, which is often intractable in high-dimensional settings. This property makes it particularly well-suited for generative modeling.

The goal of score-based learning is to approximate the true score function using a parameterized model $s_\theta(x)$. Ideally, this is done by minimizing the expected squared error between the model and the true score:

$$J(\theta) = \frac{1}{2} \int \|s_\theta(x) - \nabla_x \log p_{\text{data}}(x)\|^2 p_{\text{data}}(x)\, dx$$

However, this objective is **intractable in practice**, for two key reasons:

1. The true score function $\nabla_x logp_{data}(x)$ is unknown.
2. The data distribution $p_{data}(x)$ is only available through finite samples—often modeled as a sum of Dirac delta functions—making the gradient ill-defined and unsuitable for direct optimization.

To circumvent this, researchers turn to **score matching** methods, such as **denoising score matching** (DSM), which reformulate the objective in a tractable way by learning to predict the score of noisy versions of the data.

## 2.2. Annealed Langevin Dynamics

**Langevin dynamics** is a Markov Chain Monte Carlo (MCMC) technique that generates samples from a distribution using its score function. The basic update rule is:

$$x_{t+1} = x_t + \frac{\epsilon}{2}\nabla_x \log p(x_t) + \sqrt{\epsilon} z_t, \quad z_t \sim \mathcal{N}(0, I)$$

Here, $\epsilon$ is a small step size and $z_t$ is Gaussian noise. Intuitively, the score term nudges samples toward high-density regions, while the noise ensures stochasticity and exploration.

In high-dimensional data, however, **standard Langevin dynamics** struggles with *slow mixing*—especially when navigating between isolated modes of the distribution. To address this, **Annealed Langevin Dynamics (ALD)** introduces a series of decreasing noise levels, allowing the sampler to explore the space more effectively:

- **Early stages** use large noise to make wide exploratory jumps, helping the sampler overcome low-density barriers between modes.
- **Later stages** use small noise to refine samples and converge precisely to high-probability regions.

This gradual reduction in noise is called **annealing**, and it is typically implemented using a geometric progression of noise scales.

To make this work in practice, models are trained to estimate **noise-conditional score functions**: that is, they learn to predict the score of data perturbed by Gaussian noise at various scales. This is critical because at high noise levels, samples reside in low-density areas where the original score function is poorly defined. Training across multiple noise levels ensures the model can guide sampling even in these difficult regions.

Practical considerations for annealed Langevin implementations include:

- Using 10–20 noise scales logarithmically spaced from a high initial noise (e.g., covering the data's diameter) to a small final noise.
- Applying **Exponential Moving Average (EMA)** to model weights during sampling to stabilize training and improve sample quality.

This annealing strategy, combined with noise-conditional training, allows score-based models to generate high-fidelity samples via a principled, probabilistic framework.

# 3. Advances Beyond Naive Score-Based Modeling

In previous sections, we described how score-based generative models are trained using score matching and sampled using Langevin dynamics. While conceptually elegant, this **naive**

**approach** encounters several practical limitations. This section outlines the pitfalls of basic score-matching strategies and introduces robust improvements that rely on noise perturbation and continuous-time stochastic differential equations (SDEs).

### 3.1. Naive Score-Based Modeling and Its Pitfalls

The most straightforward pipeline—training a model to estimate the score function from clean data, and then sampling via Langevin dynamics—has not historically yielded competitive generative performance. This is largely due to the **sparsity of the data distribution** in high-dimensional spaces. In practice:

- The true data distribution $p_{data}(x)$ is concentrated in a small region of the input space.
- Outside this region, the density drops sharply, and the score function becomes poorly defined or hard to learn.
- Because the model rarely observes these low-density areas during training, it generalizes poorly when Langevin dynamics traverses them during sampling.

As a result, naive score-based models often produce **low-quality or noisy samples**, suffer from **divergent trajectories**, or **fail to mix** between modes.

### 3.2. Noise-Conditional Score Models via Data Perturbation

To mitigate the poor generalization in low-density regions, score-based models are trained not on the clean data directly, but on **perturbed versions of the data**. Specifically, Gaussian noise is added to each data point:

$$\tilde{x} = x + \sigma z, \quad z \sim \mathcal{N}(0, I)$$

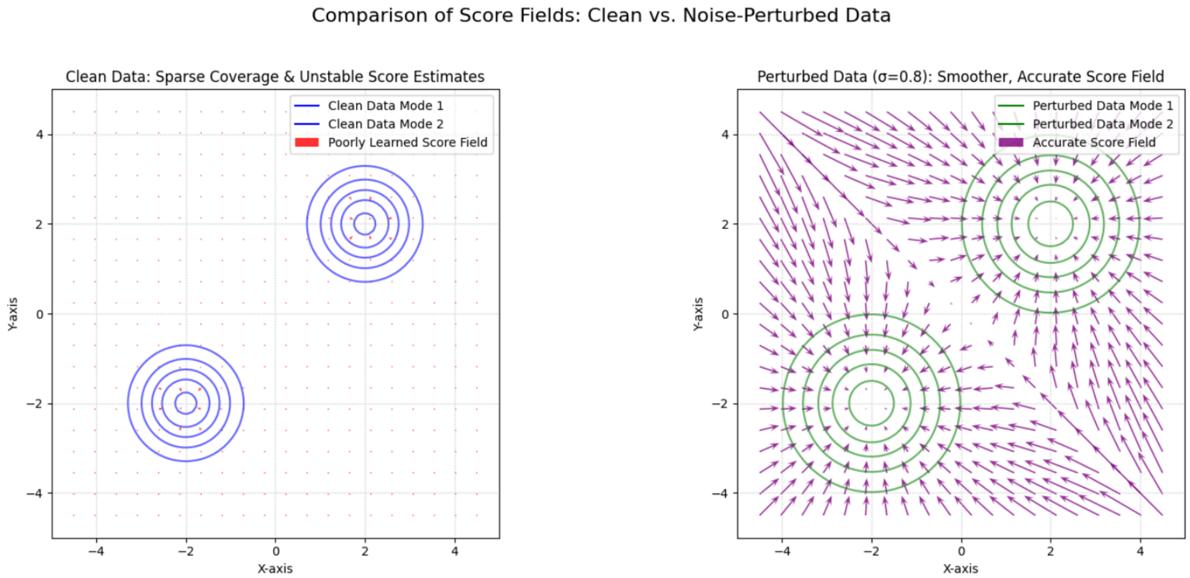
By varying the noise scale $\sigma$, the data distribution is “blurred,” effectively populating low-density regions and making score estimation easier across a wider domain.

This leads to the concept of **noise-conditional score networks**—models trained to predict the score of $p(x\sim|\sigma)$ for various noise levels $\sigma$. This idea also underpins **denoising score matching**, where the model learns to recover clean data from its noisy counterpart.

Key benefits include:

- **Improved coverage** of the input space.
- **Robust score learning** even in ambiguous or sparse regions.

- A **natural annealing schedule** during sampling, where noise levels are gradually decreased from high to low.




The noise effectively creates a family of smoother distributions that are easier to model. As the model learns score functions for multiple noise scales, it becomes capable of guiding samples from randomness to data, even when the initial points are far from the data manifold.

### 3.3. Score-Based Generative Modeling via Stochastic Differential Equations

While the multi-scale noise strategy improves modeling significantly, it still uses a **discrete sequence** of noise levels. This idea can be generalized further: by considering **infinitely many noise levels**, we can frame the generative process as solving a **Stochastic Differential Equation (SDE)**.

Formally, we define a **forward SDE** that gradually transforms the data distribution into a simple noise distribution (e.g., Gaussian) over time $t \in [0, T]$. For example, a **variance-exploding SDE** might take the form:

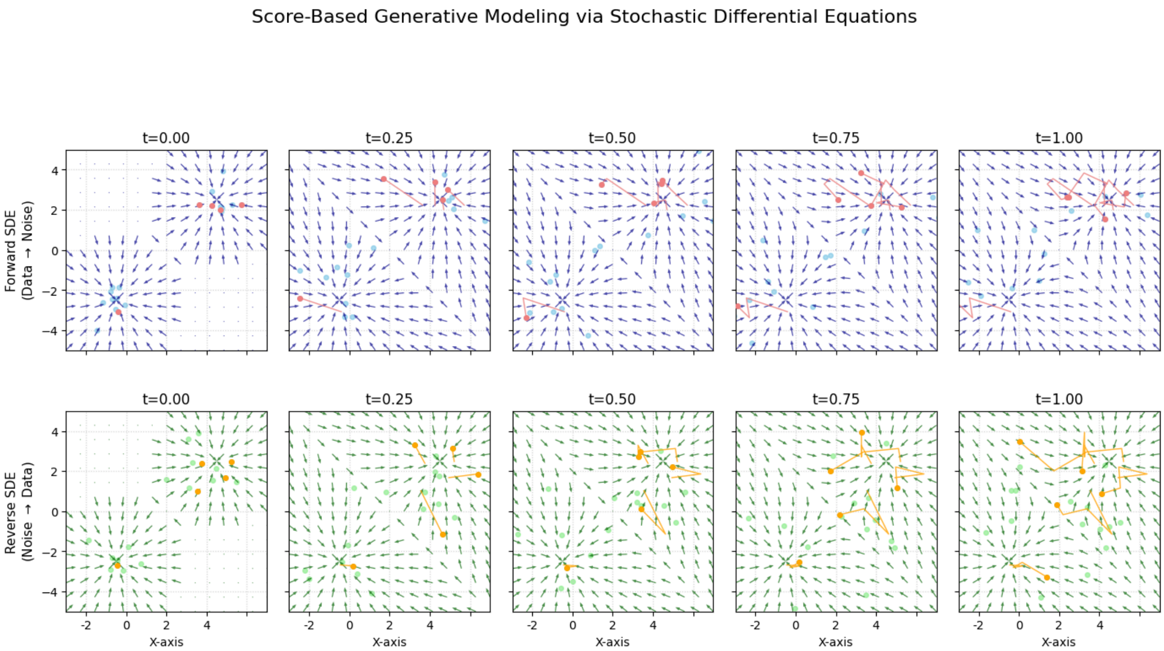
$$dx = \sqrt{g(t)}\, dw$$

where $w$ is a standard Wiener process and $g(t)$ controls the diffusion rate.

The model then learns the **score function** $s_\theta(x, t)$ of this time-dependent noisy distribution and uses it to define a **reverse-time SDE** that transforms noise back into data:

$$dx = [\text{drift}]\, dt + [\text{score-guided correction}]\, dt$$

Sampling from this reverse SDE (e.g., via Euler–Maruyama or Predictor–Corrector solvers) yields high-quality samples. Furthermore, this continuous-time framework enables **exact log-likelihood computation**, **controllable generation**, and **unified interpretations** of diffusion models, normalizing flows, and denoising autoencoders.




Across a timeline from clean data at $t = 0$ to pure noise at $t = T$, the **top row** illustrates the **forward SDE**, showing how an initial two-mode data distribution (sky-blue samples) gradually diffuses into a broad Gaussian noise distribution under the influence of increasing time-dependent variance. The overlaid dark blue vector fields represent the score function of the diffusing data, which points towards regions of higher density at each time step. Conversely, the **bottom row** demonstrates the **reverse SDE**, where samples starting from pure noise (light green) are progressively transformed back into the clean, multi-modal data distribution as time effectively runs backward (or the reverse SDE evolves forward). The dark green vector fields, identical to those above, highlight how the learned score function $s_\theta(x, t)$ precisely guides these noisy samples towards the true data manifold, demonstrating the generative power derived from modeling continuous-time diffusion.

## 4. Conclusion and Forecast

### 4.1. Final remarks

Score-based generative modeling has emerged as a powerful and theoretically grounded alternative to traditional generative paradigms. By shifting focus from directly modeling the data distribution to estimating its score function, these models offer a flexible framework that overcomes many of the limitations seen in likelihood-based models (e.g., architectural constraints) and adversarial training (e.g., instability and mode collapse).

Through the incorporation of **multi-scale noise perturbation** and **stochastic differential equations (SDEs)**, score-based methods unify ideas from various fields:

- From **diffusion processes**, they inherit a natural forward–reverse sampling scheme and a noise-driven perspective on data generation;

- From **denoising autoencoders**, they borrow the intuition of recovering clean data from noisy observations;

- From **normalizing flows**, they gain the ability to compute exact likelihoods in a continuous-time setting via probability flow ODEs.

This unification leads to a generative framework that is not only **theoretically elegant** but also **empirically competitive**, now achieving state-of-the-art results in image synthesis and offering new tools for controllable generation and inverse problem solving.

### 4.2. Open Challenges and Future Directions

Despite its promise, score-based generative modeling still faces several challenges that offer opportunities for further research:

- **Scalability and Memory**: Training high-resolution models (e.g., for 1024×1024 images) or video requires substantial computational resources. Future work could explore **memory-efficient architectures**, **adaptive noise schedules**, or **multi-resolution training strategies**.

- **Theoretical Guarantees**: While score-based models are empirically effective, formal guarantees around convergence, sample quality bounds, and generalization remain limited. More rigorous **theoretical analysis** could improve understanding and inform better algorithm design.

- **Applications Beyond Generation**: The framework shows strong promise in **inverse problems** such as denoising, inpainting, super-resolution, and even scientific imaging (e.g., MRI reconstruction). Extending score-based modeling into **domain-specific**

**settings** (e.g., genomics, robotics, simulation) could unlock new cross-disciplinary impact.

Score-based generative modeling represents not just an incremental improvement over existing methods, but a conceptual shift in how we approach generative learning. As the community continues to refine its mathematical tools, optimize its computational strategies, and broaden its applications, this class of models is poised to become a foundational pillar in the next era of generative AI.

## References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